

Dino Jump Code Reading Activity

Read the code carefully and answer the following questions without modifying the code.

Understanding Variables

1. What does `WIDTH` represent in this game?
2. What does `GROUND_Y` represent?
3. Why do we store `FPS` in a variable instead of writing `60` directly everywhere?
4. What is the purpose of `velocity_y`?
5. What is the difference between `dino_y` and `velocity_y`?
6. Why is `jump_strength` a negative number?
7. What happens if we change:

```
gravity = 1
```

to:

```
gravity = 3
```

Understanding Functions

8. What is the purpose of the `draw_text()` function?
9. Why do we pass `text`, `x`, and `y` as parameters?
10. Which functions are called from the main menu?
11. What does the `return` statement do inside the `credits()` function?
12. What would happen if we removed:

```
pygame.display.update()
```

Understanding the Menu

13. Which key starts Easy mode?
14. Which key starts Hard mode?
15. Which key opens Credits?
16. Why does the menu use:

```
while True:
```

instead of running only once?

17. What happens when the player presses ESC in the menu?

Predict the Behavior

18. What will happen if we change:

```
obstacle_speed = 6
```

to:

```
obstacle_speed = 20
```

19. What will happen if we change:

```
jump_strength = -18
```

to:

```
jump_strength = -8
```

20. What will happen if we change:

```
jump_strength = -30
```

21. What will happen if we remove:

```
on_ground = True
```

from the landing code?

22. What will happen if we remove:

```
obstacle_x -= obstacle_speed
```

Understanding the Game Loop

23. Why is there a loop:

```
while running:  
    inside game_loop()?
```

24. What would happen if we removed:

```
clock.tick(FPS)
```

25. Why do we process events every frame?

26. What happens if the player closes the window?

27. Why do we check:

```
if not game_over:  
    before updating the game?
```

Collision Detection

28. What is stored inside `dino_rect`?

29. What is stored inside `obstacle_rect`?

30. What does this line do?

```
dino_rect.collidect(obstacle_rect)
```

31. What happens when a collision is detected?

32. Why is collision detection easier with rectangles?

Score System

33. When does the score increase?

34. Why does the score not increase every frame?

35. What happens when the obstacle leaves the screen?

36. Explain this line:

```
obstacle_x = WIDTH + random.randint(100, 400)
```

37. Why was `random.randint()` used here?

Drawing

- 38. What draws the ground?
- 39. What draws the dinosaur?
- 40. What draws the obstacle?
- 41. Why is the dinosaur drawn every frame instead of only once?

Modification Questions

- 42. Which line would you modify to make the dinosaur larger?
- 43. Which variable would you modify to make obstacles taller?
- 44. How would you make the game start with 3 obstacles instead of 1?
- 45. Where would you add sound effects for jumping?
- 46. Where would you add a background image?
- 47. How would you make Hard mode more difficult than Easy mode?
- 48. How would you add a second type of obstacle?
- 49. How would you display the high score?
- 50. If you could add one feature to this game, what would it be and where in the code would you implement it?